

adenc-beamer.sty Feature Showcase

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May 7, 2026

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Theorem Environments

Theorem Environments

Assumption

The matrix $A \in \mathbb{R}^{n \times n}$ is symmetric and has a simple largest eigenvalue λ_1 .

Claim

For any unit vector x , we have $x^\top Ax \leq \lambda_1$.

Proof.

We first prove the following claim. ... □

Remark

The converse of Lagrange's theorem fails in general.

Observation

If x is an eigenvector of A , then any scalar multiple of x is also an eigenvector.

Math Commands

Number Sets & Blackboard Bold

Number sets (long and short forms):

$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{F}, \mathbb{P}$

equivalent to `\bN` , `\bZ` , `\bQ` , `\bR` , `\bC`
 , `\bF` , `\bP` .

Mathcal shortcuts `\cA` – `\cZ` :

$\mathcal{A}, \mathcal{B}, \mathcal{F}, \mathcal{G}, \mathcal{H}, \mathcal{L}, \mathcal{M}, \mathcal{X}$

Mathscr shortcuts `\sA` – `\sZ` :

$\mathscr{A}, \mathscr{B}, \mathscr{F}, \mathscr{L}, \mathscr{M}$

Imaginary unit: `i` (`\ii`)

Indicator function: $\mathbb{1}\{A\}$ (`\ind {A}`)

Delimiters & Operators

Auto-sizing delimiters:

$$\left| \frac{x}{y} \right|, \quad \left\| \frac{v}{w} \right\|, \quad \left\lceil \frac{n}{2} \right\rceil, \quad \left\lfloor \frac{n}{2} \right\rfloor$$

Vectors & matrices:

$$\mathbf{v}, \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Calculus:

$$Df, \quad \int_0^1 f(x) \, dx, \quad \int e^{it} \, dt$$

Probability:

$$\mathbb{E}[X], \quad \text{Var}(X), \quad \text{Cov}(X, Y), \quad X \perp\!\!\!\perp Y$$

Annotations & Markup

Draft Annotations

`\draftnote [Label][color]{text}` — inline draft comment.

`\draftnote {fix this}` produces **Note: fix this**

`\draftnote [TODO][blue]{rewrite}` produces **TODO: rewrite**

`\vocab {word}` — bold vocabulary term, as in: a **normal subgroup** $N \trianglelefteq G$ satisfies $gNg^{-1} = N$.

Two-Column Layout

`\twocolumns {lw}{rw}{left content}{right content}` splits a frame:

Left column ($lw=0.45$)

Content here spans 45% of `\textwidth`

.

Item one

Item two

Right column ($rw=0.45$)

Content here spans another 45%.

(i) First

(ii) Second